TELECOM CUSTOMER CHURN ANALYSIS USING EXCEL DASHBOARD

1. Introduction

Customer churn is a critical business challenge in the telecommunications industry, as retaining existing customers is significantly more cost-effective than acquiring new ones. This project aims to analyze customer churn behavior using Excel-based data visualization techniques to identify patterns, high-risk customer segments, and key business factors associated with churn.

Through interactive dashboards, pivot tables, KPI cards, and visual analytics, the study explores how demographic, contractual, financial, and service-related variables influence customer retention.

2. Problem Statement

Telecom companies often face revenue losses due to customer attrition. The objective of this project is to analyze customer behavior and identify the major factors contributing to churn, enabling the business to make informed retention and pricing decisions.

3. Dataset Description

The dataset used for this analysis contains telecom customer records with service usage, billing, contract, and churn information.

Dataset Overview

Attribute	Description
CustomerID	Unique customer identifier
Gender	Male/Female
SeniorCitizen	Senior citizen status
Tenure	Number of months with company
MonthlyCharges	Monthly billing amount
TotalCharges	Lifetime charges accumulated
Contract	Contract type
PaymentMethod	Billing/payment mode
InternetService	Type of internet service
TechSupport	Availability of technical support

Attribute	Description
OnlineSecurity	Security add-on subscription
StreamingTV / Movies	Entertainment service subscription
Churn	Customer retention status

4. Tools and Techniques Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- KPI Cards
- Histograms
- Scatter Plots
- Pie / Donut Charts
- Bar / Column Charts

5. Key Performance Indicators (KPIs)

KPI	Value
Total Customers	7043
Churn Rate	27%
Retention Rate	73%
Average Customer Tenure	32.37 Months
Average Total Charges	$2,283.30

6. Analysis and Insights

6.1 Churn by Contract Type

Month-to-month contract customers exhibit the highest churn, while two-year contract customers show the lowest churn.

Insight:
Long-term contracts significantly improve retention.

Recommendation:
Encourage month-to-month customers to switch to annual/long-term contracts through discounts and loyalty incentives.

6.2 Distribution of Monthly Charges

Customer monthly charges show a bimodal distribution, with concentrations in low-cost and premium billing segments.

Insight:
The company serves two major pricing segments.

Recommendation:
Optimize mid-tier pricing plans to attract customers between low and premium segments.

6.3 Payment Method Analysis

Electronic check users demonstrate the highest churn among payment methods.

Insight:
Manual/less automated payment methods may correlate with higher churn.

Recommendation:
Promote automatic payment enrollment through incentives.

6.4 Tenure vs Churn

Customers with lower tenure (0–12 months) churn significantly more than long-term customers.

Insight:
New customers are most vulnerable to churn.

Recommendation:
Strengthen onboarding and early-stage retention programs.

6.5 Tenure vs Total Charges

A strong positive relationship exists between tenure and total charges.

Insight:
Longer customer retention substantially increases lifetime revenue.

Recommendation:
Retention strategies should prioritize extending customer lifetime.

6.6 Internet Service Analysis

Fiber optic customers show the highest churn among internet service users.

Insight:
Fiber optic customers may face pricing or service-quality concerns.

Recommendation:
Investigate fiber optic service satisfaction and pricing competitiveness.

6.7 Tech Support Impact

Customers without tech support exhibit significantly higher churn.

Insight:
Support services positively influence retention.

Recommendation:
Bundle tech support with internet plans.

6.8 Online Security Analysis

Customers subscribed to online security churn less than those without.

Insight:
Security add-ons increase customer stickiness.

Recommendation:
Promote bundled security offerings.

6.9 Streaming Service Analysis

Streaming TV and Streaming Movies customers display varied churn behavior.

Insight:
Entertainment services alone do not guarantee retention.

Recommendation:
Evaluate bundled pricing/value of streaming services.

6.10 Senior Citizen Analysis

Senior citizens account for a smaller portion of customers but show notable churn behavior.

Insight:
Senior customer retention may require targeted service/support strategies.

Recommendation:
Develop senior-friendly pricing/support plans.

6.11 Gender Analysis

Male and female churn patterns show minimal variation.

Insight:
Gender is not a major churn determinant.

Recommendation:
Retention strategies should prioritize behavioral/service factors over gender segmentation.

6.12 Add-On Services Analysis

Customers with more service add-ons demonstrate lower churn.

Insight:
Bundled services improve customer retention.

Recommendation:
Promote bundled packages and upselling strategies.

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7. Overall Findings

Major churn drivers identified include:

1. Month-to-month contracts
2. Low customer tenure
3. Fiber optic internet service
4. Lack of tech support / online security
5. Electronic check payment method

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8. Business Recommendations

1. Promote long-term contracts with incentives
2. Improve onboarding for new customers

3. Investigate fiber optic dissatisfaction
4. Encourage bundled service adoption
5. Incentivize automatic payment methods
6. Offer retention campaigns for high-risk segments

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9. Conclusion

This analysis demonstrates that customer churn is strongly influenced by contract type, tenure, service subscriptions, billing behavior, and support services. Through Excel-based dashboarding and visualization, actionable business insights were extracted to support telecom retention strategies.

By targeting high-risk customer segments and improving service/value offerings, telecom providers can reduce churn and improve customer lifetime value.